

Decentralized Computing and Blockchains

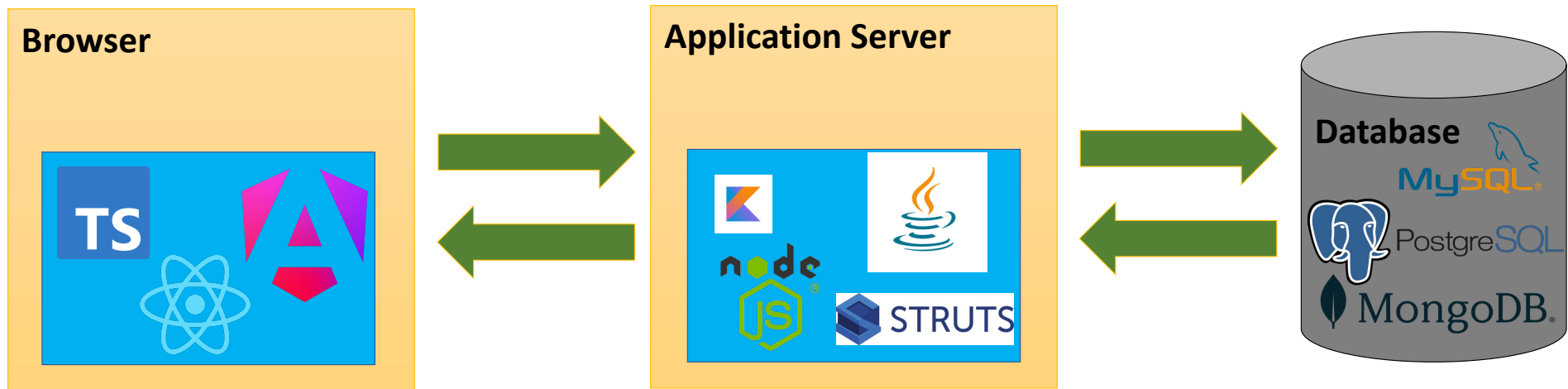
Decentralized/Web3 Applications

(general ideas and pointers)

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DI-FCUL

Typical Web Application



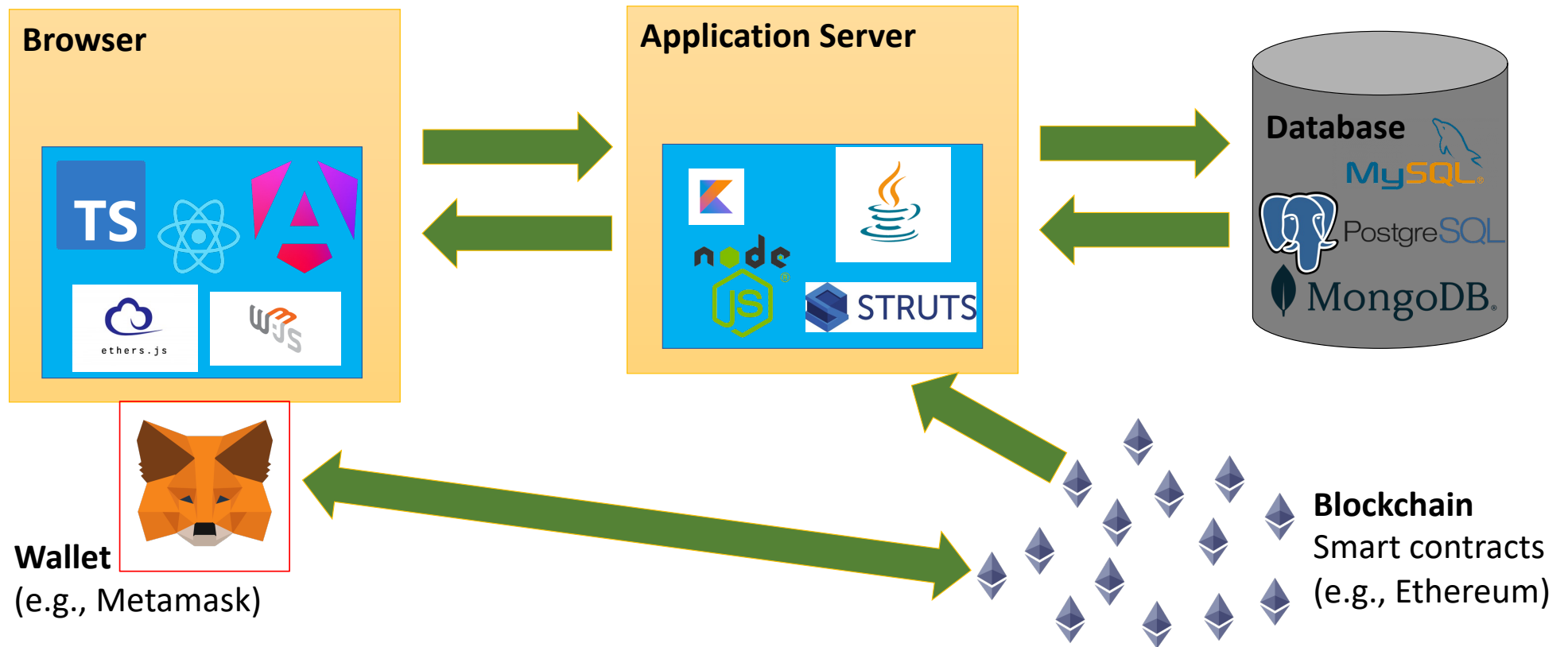
Front end:

- Language: Javascript, Typescript, etc.
- Frameworks: React, Angular, etc.

Back end:

- Language: Typescript, Java, Kotlin, etc.
- Frameworks: Struts, Node.js, etc.

Typical Web3 Application



Development Environment

- VSCode
 - Plugin Solidity by Nomic Foundation
- Node.js
 - Typescript language (Javascript with types)
 - <https://nodejs.org/en/download>
- Hardhat Framework
 - Allows the local development of Solidity smart contracts
 - <https://hardhat.org/docs/tutorial>
- Ethers.js (integrated with hardhat)
 - Library for accessing wallet
 - <https://hardhat.org/docs/plugins/hardhat-ethers>
 - In the [HelloEthers.zip](#) file in Moodle I added a script to access the local Ethereum node using this lib
- Metamask
 - Wallet for keeping your accounts (I recommend the chrome plugin)
 - Use Ethers.js to connect js with metamask, e.g., <https://dev.to/drunknells/simple-start-to-smart-contract-development-with-hardhat-4c1h>
 - In the [HelloEthers.zip](#) file in Moodle I added a variation of this tutorial webpage to access the Counter contract of Hardhat tutorial using Ethers.js and Metamask